

Economic Order Quantity Model (Answer Key)

Task 1:

ABI GmbH operates two warehouses. Each has the same stable demand of 50 units per year, annual per-unit holding costs of €20, and per-order shipment costs of €50.

- a. What are the annual logistics costs (holding + shipment)?

Answer: € 632.46

- b. The company has an opportunity to pool all demand and operate only one warehouse to serve all customers. How much would ABI GmbH save by pooling warehouses?

Answer: € 185.24; 29.29%

Task 2:

Tek Pak Inc. manufactures beers. The firm has recently bought a huge warehouse and has ample capacity for inventory. The holding cost of a beer crate is €0.65 per crate per year; each delivery causes a setup cost of €25. Beers have a flat demand of 130 crates per year. Tek Pak has an initial inventory of 100 crates. The replenishment lead time from the manufacturing plant to the warehouse for the order is 2 weeks.

- a. Find optimal order quantity.

Answer: 100

- b. What is the inventory level to place new orders for beers?

Answer: 5

- c. After how many weeks should you place the first order?

Answer: 38 weeks

Task 3:

The setup cost of ordering kerosene oil is €10 per order, the annual holding cost of kerosene is €2 per gallon, and the annual demand rate is 8,000 gallons.

- a. Assume an infinite planning horizon.

- i. How many units should you order each time?

Answer: 282.84

- ii. How many order(s) should you place per year?

Answer: 28.28

- iii. How often would you order, on average, in a 5-weeks time window?

Answer: 2.72 times

- b. Now, assume your entire selling season is 5 weeks, with everything else remaining the same.

- i. How often should you order?

Answer: 3 times

- ii. How many items should you order each time?

Answer: 256.41

- c. Calculate the average total costs for 5 weeks windows and for an infinite window. Which window period has higher average total costs, by how much percent, and why?

Answer: The total cost is 0.48% larger in a finite time period compared to an infinite period as the order frequency in a finite period (3 times) is rounded from the optimum order frequency (2.72 times), and total costs in the finite period are, thus, calculated using rounded values.